

PATIENT HEALTH MONITORING SYSTEM

(MINI PROJECT REPORT)

Arduino-Based Health Monitoring Project

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(2026)

ABSTRACT

Patient Health Monitoring System is an Arduino-based healthcare monitoring project designed to monitor a patient's body temperature and heart rate in real time. The system uses an LM35 temperature sensor and a heart rate sensor to collect physiological data. The Arduino Uno processes the sensor readings and continuously monitors the patient's condition. If the temperature or heart rate exceeds the predefined threshold values, an LED and buzzer are activated to alert caregivers about the abnormal condition. The system provides a simple, low-cost, and effective solution for basic health monitoring and can be further enhanced with wireless communication and cloud-based data storage for remote patient monitoring applications.

INTRODUCTION

Healthcare monitoring plays a crucial role in ensuring the well-being of patients by continuously tracking vital health parameters. Early detection of abnormal conditions such as high body temperature and irregular heart rate can help prevent serious health complications and improve patient care. With the advancement of embedded systems and sensor technologies, low-cost and efficient health monitoring systems can be developed for real-time observation of patient health.

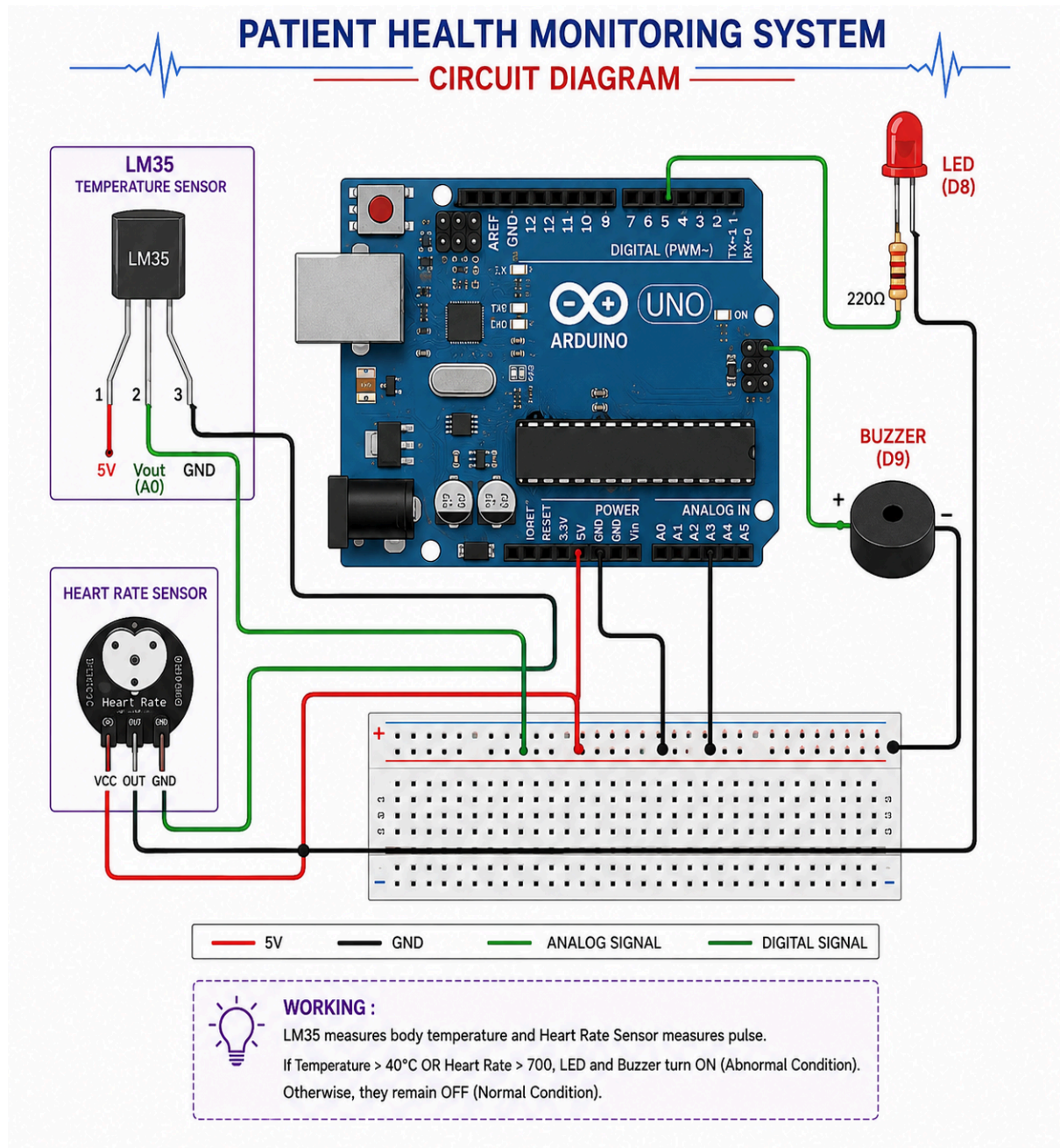
The Patient Health Monitoring System is an Arduino-based project designed to monitor two important physiological parameters: body temperature and heart rate. The system uses an LM35 temperature sensor to measure body temperature and a heart rate sensor to detect pulse signals. The sensor data is collected and processed by the Arduino Uno microcontroller, which continuously evaluates the patient's condition.

This project demonstrates the practical application of sensors, microcontrollers, and embedded programming in healthcare monitoring. It provides a simple, cost-effective, and user-friendly solution for basic patient monitoring and serves as a foundation for more advanced healthcare systems. Future enhancements may include wireless communication, cloud-based data storage, mobile notifications, and real-time remote monitoring to improve accessibility and efficiency in healthcare services.

COMPONENTS USED

S:NO	COMPONENT	QUANTITY	PURPOSE
1.	Arduino UNO	1	Microcontroller used to process sensor data
2.	Temperature Sensor(LM35)	1	Measures Body Temperature
3.	Heart Rate Sensor	1	Monitors Pulse/Heart rate
4.	LED	1	Visual Alert Indicator
5.	BUZZER	1	Audio Alert Indicator
6.	Breadboard	1	Circuit Prototyping
7.	Jumper Wires	As required	Electrical Connection
8.	USB Cable	1	Power Supply & Code Uploading

CIRCUIT DIAGRAM



DESCRIPTION

The Patient Health Monitoring System is an Arduino-based embedded system designed to monitor a patient's vital health parameters in real time. The system measures body temperature using an LM35 temperature sensor and monitors pulse activity using a heart rate sensor. These sensors continuously provide data to the Arduino Uno microcontroller for processing and analysis.

The Arduino reads the sensor values through its analog input pins and compares the readings with predefined threshold values. If the temperature or heart rate exceeds the safe limit, the system identifies the condition as abnormal. In such cases, an LED and a buzzer are activated to provide immediate visual and audible alerts. If the readings remain within the normal range, the alert devices remain turned off.

The monitored values and patient status are displayed through the Arduino Serial Monitor, allowing users to observe the sensor readings in real time. This enables quick identification of potential health issues and facilitates timely response.

The project demonstrates the integration of sensors, microcontrollers, and alert mechanisms to create a simple and cost-effective healthcare monitoring solution. It can be used for educational purposes, prototype healthcare applications, and basic patient monitoring systems. Future enhancements may include wireless communication, cloud storage, mobile notifications, and advanced health analytics for remote patient monitoring.

WORKING PRINCIPLE

The Patient Health Monitoring System operates by continuously monitoring a patient's body temperature and heart rate using sensors connected to an Arduino Uno microcontroller. The LM35 temperature sensor measures the body temperature and produces an analog voltage proportional to the detected temperature. Similarly, the heart rate sensor generates analog signals corresponding to the patient's pulse activity.

The Arduino Uno reads the analog signals from both sensors through its analog input pins (A0 and A1). The acquired sensor values are processed and compared with predefined threshold values programmed in the Arduino code.

If the temperature value exceeds the specified threshold or the heart rate sensor value rises above the predefined limit, the system considers the patient's condition abnormal. In response, the Arduino activates an LED and a buzzer to provide immediate visual and audible alerts. This helps in quickly identifying potential health risks and enables timely intervention.

When both sensor readings remain within the normal range, the LED and buzzer remain turned off, indicating that the patient's condition is normal. The temperature, heart rate readings, and patient status are continuously displayed on the Arduino Serial Monitor, allowing real-time observation and analysis.

Thus, the system provides a simple, low-cost, and effective method for monitoring essential health parameters and generating alerts whenever abnormal conditions are detected.

ARDUINO IDE CODE

```
#define TEMP_SENSOR A0
#define HEART_SENSOR A1
#define LED 8
#define BUZZER 9

void setup() {
  Serial.begin(9600);
  pinMode(LED,OUTPUT);
  pinMode(BUZZER,OUTPUT);
}

void loop() {
  int TempValue = analogRead(TEMP_SENSOR);

  int PulseValue = analogRead(HEART_SENSOR);

  Serial.print("Temperature = ");
  Serial.print(TempValue);
  Serial.print(" c");

  Serial.print("Heart Rate: ");
  Serial.print(PulseValue);

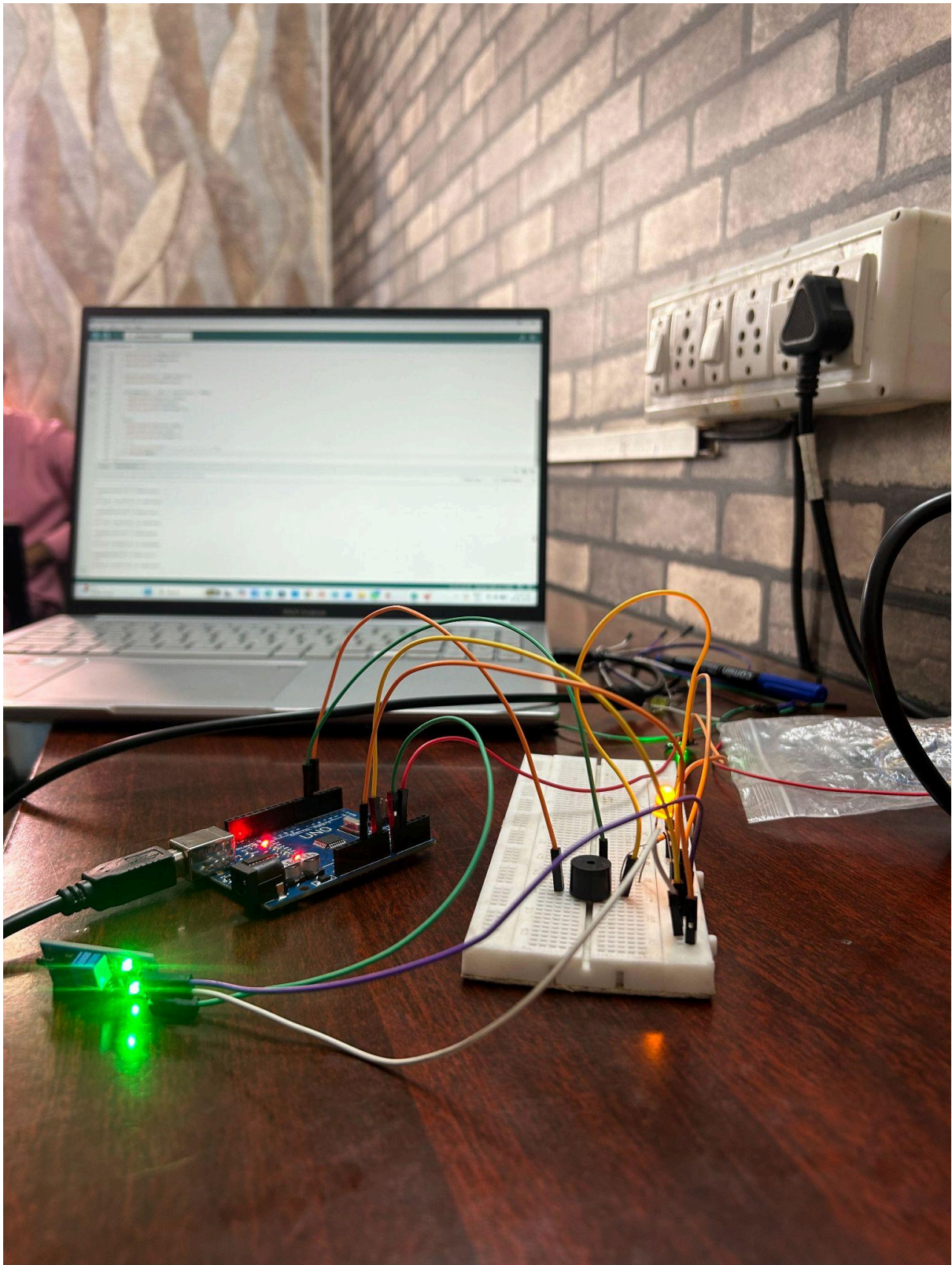
  if(TempValue>40 || PulseValue>700)
  {
    digitalWrite(LED,HIGH);
    digitalWrite(BUZZER,HIGH);

    Serial.println("PATIENT CONDITION ABNORMAL");

  }
  else{
    digitalWrite(LED,LOW);
    digitalWrite(BUZZER,LOW);

    Serial.println("PATIENT CONDITION NORMAL");
  }
  delay(1000);
}
```


RESULT & OUTPUT




```
23. digitalWrite(buzzer,HIGH);
24     digitalWrite(LED,HIGH);
25     Serial.println("abnormal");
26 }
27 else{
28     digitalWrite(buzzer,LOW);
29     digitalWrite(LED,LOW);
30     Serial.println("normal");
31 }
32     Serial.println(".....");
33     delay(3000);
34 }
```

Output Serial Monitor x

Not connected. Select a board and a port to connect automatic

```
temperature:396 c heartrate:
384
normal
.....
temperature:399 c heartrate:
154
normal
.....
temperature:411 c heartrate:
188
normal
.....
temperature:414 c heartrate:
509
normal
.....
temperature:415 c heartrate:
...
```

1 31°C
Mostly cloudy



Search

FUTURE SCOPE

The Patient Health Monitoring System can be further enhanced by incorporating advanced technologies and additional healthcare features. Wireless communication modules such as Wi-Fi, Bluetooth, or GSM can be integrated to enable remote monitoring and real-time notifications. The system can also be connected to cloud platforms for storing and analyzing patient health data over extended periods.

Additional sensors such as blood oxygen (SpO_2), blood pressure, ECG, and respiration sensors can be added to improve the accuracy and functionality of the monitoring system. An LCD or OLED display can be incorporated to display health parameters directly on the device. Mobile applications can also be developed to provide users and healthcare professionals with convenient access to patient data and alerts.

These enhancements would transform the project into a comprehensive smart healthcare monitoring system suitable for home care, hospitals, and remote patient monitoring applications.

CONCLUSION

The Patient Health Monitoring System was successfully developed using an Arduino Uno, LM35 temperature sensor, heart rate sensor, LED, and buzzer. The system continuously monitors the patient's body temperature and heart rate and provides immediate alerts whenever abnormal conditions are detected.

The project demonstrates the effective integration of sensors and embedded systems for healthcare applications. It provides a simple, reliable, and cost-effective solution for basic patient monitoring. The alert mechanism helps in identifying potential health issues at an early stage, enabling timely action and improving patient safety.

Overall, the project serves as a practical implementation of embedded system concepts and highlights the potential of sensor-based healthcare monitoring systems in modern medical applications.